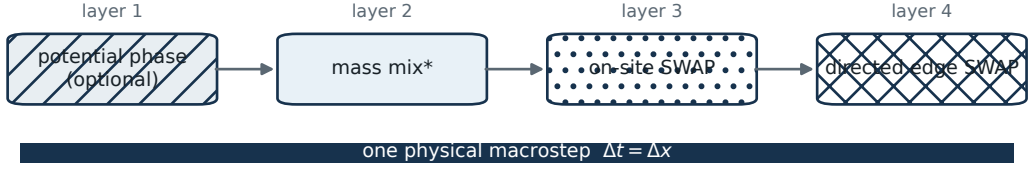
**B**

## Product walk: declared ordered decomposition

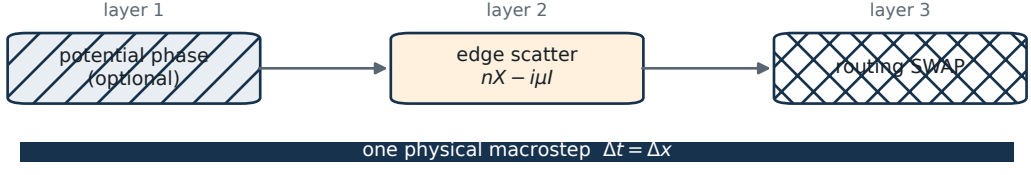
$U_{\mathrm{PW}} = S\,C_m\,V$; $*C_m$ frozen only at one excitation; $|00\rangle, |11\rangle$ phases unfrozen




free depth 3 | active depth 4 | gate-work $3N$ / $4N$ | storage $2N$ | ancillas 0 | radius 1

## Direct QCA: registered edge-scatter/routing clock

$U_{\mathrm{D}} = K^{1/2} C_{\arcsin \mu} K^{1/2}$; registered edge-scatter/routing clock shown




free depth 2 | active depth 3 | gate-work $2N$ / $3N$ | storage $2N$ | ancillas 0 | radius 1